Hossein Baktash

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Education

2021 – now **Ph.D. in Electrical and Computer Engineering**, *Carnegie Mellon University*, Pittsburgh, PA, *advisor:* Keenan Crane.

2015–2020 **B.Sc. in Computer Engineering**, *Sharif University of Technology*, Tehran, Iran, *GPA: 17.01/20.*
- *Minor in Mathematics*

2014–2015 **Team selection camp for International Mathematical Olympiad**, *Young Scholars Club*, Tehran, Iran.

2011–2014 **Diploma in Mathematics and Physics Discipline**, *Shahid Soltani High School (NODET)*, Karaj, Iran,
GPA: 18.49/20.

Publications

Putting Rigid Bodies to Rest
Hossein Baktash, Nicholas Sharp, Qingnan Zhou, Keenan Crane, Alec Jacobson
ACM Transactions on Graphics (SIGGRAPH 2025).

Computational Imaging using Ultrasonically-Sculpted Virtual Lenses
Hossein Baktash, Yash Belhe, Matteo Giuseppe Scopelliti, Yi Hua, Aswin Sankaranarayanan, Maysam Chamanzar
IEEE Intl. Conf. Computational Photography (ICCP 2022).

Some Results on Dominating Induced Matching
S. Akbari, A. Bahjati, H. Baktash, A. Behmaram, M. Roghani
Graphs and Combinatorics, 2022.

Awards and Honors

2021 **Carnegie Institute of Technology Dean's Fellowship**, *Carnegie Mellon University*.

2014 **Iranian National Mathematics Olympiad**, ***Gold medal***, Young Scholars Club, Tehran, Iran.

Research Experience

summer 2023 & 2024 **Research Scientist Intern**, *Adobe*, NYC.
Supervised by Alec Jacobson and Qingnan Zhou. Developed an algorithm to identify resting probabilities of rigid bodies and used it for inverse design of shapes with irregular dice behavior.

2022–now **Graduate Researcher**, *Geometry Collective*, Carnegie Mellon University.
Advised by Keenan Crane. Working on various novel discrete representations for shapes used in geometry processing.

2021–2022 **Graduate Researcher**, *Chamanzar Lab*, Carnegie Mellon University.
Developed an image processing pipeline to relay and deblur images using unconventional ultrasonic lenses in water.

2019–2020 **Bachelor Thesis**, *Sharif University of Technology, Tehran*.
Worked on recovering the structure of Ising Blockmodels with more than two blocks. Improved the immediate sample complexity of $\mathcal{O}(n^2 \log(n))$ to the information theoretic lower bound of $\mathcal{O}(n \log(n))$ using a variant of the max-cut SDP relaxation.

summer 2019 **Research Internship**, *INRIA - I3S Laboratoire, France*.
Studied several pruning and hashing methods to compress neural networks including HashedNet, OBD, and magnitude-based pruning with $\ell 1$ and $\ell 2$ regularization methods.

Talks

Computational Imaging using Ultrasonic Lenses
ICCP 2022, Caltech, Pasadena, CA

Teaching Experiences

Teaching Assistant.

- CMU: Discrete Differential Geometry, Mathematical Foundations of Electrical Engineering
- SUT: Linear Algebra, Probability and Statistics, Theory of Languages and Automata

Tutoring.

- Geometry, Combinatorics, Number Theory for Math/Informatics Olympiad participants

Skills

programming **C++**, **Python**, MATLAB, C, Java

Tools Blender, Illustrator, Git, Geogebra, Slurm

Languages Persian, English, Azeri

Additional Activities

Sports Mountaineering, Horse Riding, Snooker/Pool

Hobbies Drawing, Cooking

Executive experience

- **Member of Sharif University's Mountaineering Group** (2016–2020)
  Assigned as team leader several times and successfully carried out the responsibilities, which often involved leading a team of 20-30 members in nature, outdoors, and remote regions of the country.
- **Team guide in IOI 2017 (International Olympiad in Informatics)**
  I was honored to be the guide of team Latvia in the competition. Providing real-time translation and helping the team enjoy their one-week stay in Tehran.